

Undergraduate Topics in Mathematics 5
Abstract Algebra

Nathan Warner



**Northern Illinois
University**

Computer Science
Northern Illinois University
United States

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Combinatorics

1.1 Introduction

- **What is combinatorics?:** Combinatorics is a collection of techniques and a language for the study of finite or countably infinite discrete structures. Given a set of elements and possibly some structure on that set, typical questions are
 - Does a specific arrangement of the elements exists?
 - How many such arrangements are there?
 - What properties do these arrangements have?
 - Which one of the arrangements is maximal, minimal, or optimal according to some criterion?
- **Counting the number of subsets for a set:** Let $[n] = \{1, 2, \dots, n\}$, and let $f(n)$ be the number of subsets of $[n]$. Then $f(n) = 2^n$. For any particular subset of $[n]$, each element is either in that subset or not. Thus, to construct a subset, we have to make one of two choices for each element of $[n]$. Furthermore, these choices are independent of each other. Hence, the total number of choices, and consequently the total number of subsets is

$$\underbrace{2 \times 2 \times \dots \times 2}_n = 2^n.$$

- **Number of subsets without consecutive integers:** For a sequence $[n] = \{1, \dots, n\}$ we can count the number of subsets given by $f(n)$, that do not contain consecutive integers with the recurrence relation

$$f(n) = f(n-1) + f(n-2).$$

We consider two cases

1. n is not included in the subsets
2. n is included in the subsets. In this case, we build the subsets considering the subsequence $[n-2] = \{1, \dots, n-2\}$. Note that if we include n , we must exclude $n-1$, because $n-1$ and n are consecutive, this will become clear in the upcoming example.

Consider the sequence $[n] = \{1, 2, 3, 4\}$. By the relation above,

$$f(4) = f(3) + f(2).$$

Before we are able to compute this, we must define our base cases.

$$f(n) = \begin{cases} 3 & \text{if } n = 2 \\ 2 & \text{if } n = 1 \end{cases}.$$

If $n = 2$, we have $\{1, 2\}$, and the allowed subsets are $\emptyset, \{1\}, \{2\}$. If we have $n = 1$, the subsets are $\{\emptyset, \{1\}\}$. Thus

$$\begin{aligned} f(4) &= f(3) + f(2) = f(2) + f(1) + f(2) \\ &= 3 + 2 + 3 = 8. \end{aligned}$$

Let's explicitly break up the given sequence so we can see what's going on. In the first case, n is excluded, thus the sequence becomes $\{1, 2, 3\}$. If n is included, the sequence becomes $\{1, 2\}$, where we build the subsets of $\{1, 2\}$, and then add 4 to each one. Thus,

$$\begin{aligned}\{1, 2, 3\} + \{1, 2\} &= \{1, 2, 3\} + \emptyset + \{1\} + \{2\} \\ &= \{1, 2, 3\} + \{4\} + \{1, 4\} + \{2, 4\}.\end{aligned}$$

Since the sequence $\{1, 2, 3\}$ is not a base case, we must split this one up as well, we have

$$\begin{aligned}\{1, 2, 3\} &= \{1, 2\} + \{1\} + \{4\} + \{1, 4\} + \{2, 4\} \\ &= \emptyset + \{1\} + \{2\} + \emptyset + \{1\} + \{4\} + \{1, 4\} + \{2, 4\} \\ &= \emptyset + \{1\} + \{2\} + \{3\} + \{1, 3\} + \{4\} + \{1, 4\} + \{2, 4\}.\end{aligned}$$

Thus, we conclude all "good" subsets of $[n]$ either have n or don't have n . The ones that don't have n are exactly the "good" subsets of $[n - 1]$. The "good" subsets of $[n]$ that include n are exactly the "good" subsets of $[n - 2]$ together with n . Thus $f(n) = f(n - 1) + f(n - 2)$ ■

1.2 Induction and recurrence relations

- **Principal of Mathematical Induction:** Given an infinite sequence of propositions

$$P_1, P_2, P_3, \dots, P_n, \dots, .$$

In order to prove that all of them are true, it is enough to show two things

1. **The base case:** P_1 is true
2. **The inductive step:** For all positive integers k , if P_k is true, then so is P_{k+1}

Example: Show that

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}.$$

Base case:

$$1 = \frac{1(1+1)}{2} = \frac{2}{2} = 1.$$

Inductive step: P_k is given by

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}.$$

P_{k+1} is given by

$$1 + 2 + 3 + \dots + k + k + 1 = \frac{k+1(k+2)}{2}.$$

If $1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$, then

$$\begin{aligned} 1 + 2 + 3 + \dots + k + k + 1 &= \frac{k+1(k+2)}{2} \\ \frac{k(k+1)}{2} + k + 1 &= \frac{k+1(k+2)}{2} \\ \frac{k(k+1) + 2k + 2}{2} &= \frac{k^2 + 3k + 2}{2} \\ \frac{k^2 + 3k + 2}{2} &= \frac{k^2 + 3k + 2}{2}. \end{aligned}$$

Thus, we have showed that $P_k \implies P_{k+1}$ ■.

Note: Our aim is not to directly prove P_{k+1} , but to prove that P_k implies P_{k+1} . In the inductive step we assume P_k to be true, then show under this assumption, P_{k+1} is also true.

- **Understanding gauss's formula for the sum of the first n natural numbers:** Suppose we want to find the sum $1 + 2 + 3 + \dots + (n-1) + n$. We could have discovered the formula that we proved above by first writing the sum twice

$$\begin{array}{r} 1 + 2 + 3 + \dots + (n-1) + n \\ n + (n-1) + (n-2) + \dots + 2 + 1. \end{array}$$

The sum of the two numbers in each column is $n+1$, and there are n columns, so the total sum is $n(n+1)$, it then follows that the actual sum is $\frac{1}{2}n(n+1)$

- **Triangular numbers:** The sequence of integers

$$\begin{array}{ll}
 1 & 3 = 1 + 2 \\
 6 = 1 + 2 + 3 & \\
 10 = 1 + 2 + 3 + 4 & \\
 15 = 1 + 2 + 3 + 4 + 5 & \\
 &
 \end{array}$$

Are called *triangular numbers*. If you were to make a triangle of dots out of the sum, where the highest number is the base, the second highest is the layer on top of the base, etc, you would form a triangle.

- **Strong induction:** Given an infinite sequence of propositions

$$P_1, P_2, P_3, ..., P_n.$$

In order to demonstrate that all of them are true, it is enough to know two things.

1. **The base case:** P_1 is true
 2. **The inductive step:** For all integers $k \geq 1$, if $P_1, P_2, P_3, ..., P_k$ are true, then so is P_{k+1}
- **Pingala-fibonacci numbers:** Define a sequence of positive integers as follows: $F_0 = 0, F_1 = 1$, and for $n = 2, 3, ...$ we have

$$F_n = F_{n-2} + F_{n-1}.$$

This sequence is also known as *the fibonacci sequence*.

- **Lucas numbers:** Change the initial values on the fibonacci sequence. Let $L_0 = 2, L_1 = 1$, and $L_n = L_{n-2} + L_{n-1}$. Then, we get the *Lucas numbers*

$$2, 1, 3, 4, 7, 11, 18, 29, 47,$$

$$\mathcal{L}.$$

The Fourier Transform

- **The Fourier Transform:** The Fourier transform is a tool that rewrites a function in terms of its frequency content. Instead of describing a signal by “what value it has at each time,” the Fourier transform describes it by “how much of each frequency is present.”

For a function $f(t)$, the FT is written as

$$\hat{f}(\omega) = \mathcal{F}\{f(t)\},$$

and defined by

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt.$$

With inverse

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega t} d\omega,$$

which recovers the original function $f(t)$. Note that ω here is the angular frequency.

- **Derivatives become multiplication:** One reason Fourier transforms are so powerful is that they turn differentiation into multiplication.

$$\mathcal{F}\{f'(t)\} = i\omega \hat{f}(\omega).$$

Abstract Algebra

3.1 Relations and functions

- **Relations:** Let S and T be sets. A **relation** from S to T is a subset ρ of $S \times T$, where

$$S \times T = \{(s, t) : s \in S, t \in T\}.$$

If $s \in S$ and $t \in T$, then we write $s\rho t$ if $(s, t) \in \rho$, otherwise we write $s\not\rho t$. In particular, a relation on S is a relation from S to S .

- **Properties of relations:** Let ρ be a relation on S ,
 - **Reflexive:** ρ is reflexive if $a\rho a$ for all $a \in S$
 - **Symmetric:** ρ is symmetric if $a\rho b$ implies $b\rho a$
 - **Transitive:** ρ is transitive if $a\rho b$ and $b\rho c$ implies that $a\rho c$

Note: These properties are defined only for relations from a set to itself.

- **Equivalence relation:** An **equivalence relation** on a set S is a relation that is reflexive, symmetric, and transitive. We use $\sim$ to denote an equivalence relation.

As an example, on $\mathbb{R}$, let us say that $a \sim b$ if and only if $a + b$ is even. First, consider $a, b \in \mathbb{R}$. Observe that

$$a \sim a$$

since whether or not a is even or odd, the sum is always even. So, $\sim$ is reflexive. Next, we notice that if we assume $a \sim b$ (so $a + b$ is even), it immediately follows that $b + a$ is even. So, $b \sim a$ and $\sim$ is symmetric.

Now, let $a, b, c \in \mathbb{R}$. Assume that

$$a + b, b + c$$

are both even. Hence,

$$a + b + b + c = a + 2b + c = a + c + 2b$$

is also even. Notice that $2b$ is even, so we consider two cases, either $a + c$ is odd and $\sim$ is not transitive or $a + c$ is even. If $a + c$ is odd, then we have $a + c = 2\ell + 1$, and

$$a + c + 2b = 2\ell + 1 + 2b = 2(\ell + b) + 1,$$

which is odd. But, we said that $a + c + 2b$ is even, a contradiction. Thus, $a \sim c$ and $\sim$ is transitive.

Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation. ■

- **Equivalence class:** Let $\sim$ be an equivalence relation on a set S . If $a \in S$, then the **equivalence class** of a , denoted $[a]$, is the set $\{b \in S : a \sim b\}$
- **Partition:** Let S be a set, and let T be a set of nonempty subsets of S . We say that T is a **partition** of S if every $a \in S$ lies in exactly one set in T .
- **Theorem 1.1:** Let S be a set and $\sim$ be an equivalence relation on S . Then the equivalence classes with respect to $\sim$ form a partition of S . In particular, if $a \in S$, then $a \in [a]$, and $a \in [b]$ if and only if $[a] = [b]$.

Proof. Assume S is a set, and $\sim$ is an equivalence relation on S . To show that the equivalence relations with respect to $\sim$ form a partition of S , we must show that the equivalence relations are a set of nonempty pairwise disjoint subsets of S .

Let $a \in S$. By reflexivity of $\sim$, $a \sim a$. Thus, $a \in [a]$ and the equivalence classes are nonempty. Now, assume that

$$d \in [a] \cap [b].$$

Thus,

$$a \sim d, \quad b \sim d.$$

Now, if $c \in [a]$, then $a \sim c$. Notice that by transitivity,

$$d \sim a, a \sim c \implies c \sim d.$$

Similarly,

$$c \sim d, d \sim b \implies c \sim b.$$

Thus, $c \in [b]$. Since $c \in [a]$ implies that $c \in [b]$,

$$[a] \subseteq [b].$$

A similar argument, except starting with $c \in [b]$ yields that

$$[b] \subseteq [a].$$

Hence, if $d \in [a] \cap [b]$, it follows that $[a] = [b]$.

Therefore, if $a \in S$, then $a \in [a]$, and if $a \in [b]$, then $[b] = [a]$. Conversely, if $[a] = [b]$, then $a \in [a] = [b]$. ■

- **The partition:** So, now we know that for a equivalence relation $\sim$ on S ,

$$\sim \subseteq S \times S,$$

if $a \in S$, then the equivalence class of a is the set

$$[a] = \{b \in S : a \sim b\}.$$

Further, $b \in [b]$, and so it must be that $[b] = [a]$. In other words, the representative element is not unique for each class.

- **Functions:** Let S and T be any sets. A **function**

$$\alpha : S \rightarrow T$$

is a rule assigning to each $s \in S$ an element $\alpha(s) \in T$.

- **Requirements of a function:** For a relation $f \subseteq A \times B$ to be called a **function** from A to B , it must satisfy

1. **Existence / totality:** For all $a \in A$, there exists a $b \in B$ such that $(a, b) \in f$

$$\forall a \in A, \exists b \in B : (a, b) \in f.$$

2. **Uniqueness / well-definedness:** Each input in A has exactly one output in B

$$\forall a \in A, \forall b_1, b_2 \in B, ((a, b_1) \in f \wedge (a, b_2) \in f) \implies b_1 = b_2.$$

- **Properties of functions:** Consider a function $\alpha : S \rightarrow T$,
 - **Injective:** α is injective if $\alpha(s_1) = \alpha(s_2)$ implies $s_1 = s_2$ for all $s_1, s_2 \in S$
 - **Surjective:** α is surjective if for all $t \in T$, there exists an element $s \in S$ such that $\alpha(s) = t$
 - **Bijective:** α is bijective (invertible) if it is both injective and surjective

We call a bijective function a **bijection**, which is therefore also a **injection** and a **surjection**.

Notice that the inverse of a function only satisfies existence / uniqueness if it is both injective (uniqueness) and surjective (existence).

- **Composition:** Let R, S , and T be sets, and let $\alpha : R \rightarrow S$ and $\beta : S \rightarrow T$ be functions. Then, the **composition** $\beta \circ \alpha$, or simply $\beta\alpha$, is the function from R to T given by

$$(\beta\alpha)(r) = \beta(\alpha(r))$$

for all $r \in R$. Note that $\alpha\beta$ and $\beta\alpha$ are likely different functions.

- **Theorem 1.2: Properties of the composition:** Let $\alpha : R \rightarrow S$, $\beta : S \rightarrow T$, and $\gamma : T \rightarrow U$ be functions. Then,
 1. $(\gamma\beta)\alpha = \gamma(\beta\alpha)$
 2. If α and β are injective, then so is $\beta\alpha$
 3. If α and β are surjective, then so is $\beta\alpha$
 4. If α and β are bijective, then so is $\beta\alpha$

Proof. First, notice that

$$\begin{aligned} (\gamma\beta)\alpha &= \gamma(\beta\alpha), \\ \gamma(\beta\alpha) &= \gamma(\beta\alpha). \end{aligned}$$

Hence, the first property holds. Now, consider $r_1, r_2 \in R$, and assume that

$$\beta(\alpha(r_1)) = \beta(\alpha(r_2)).$$

Since β is injective, $\alpha(r_1) = \alpha(r_2)$, and since α is injective, $r_1 = r_2$. Thus, $\beta\alpha$ is injective.

Next, consider $t \in T$. Since β is surjective, there exists a $s \in S$ such that $\beta(s) = t$. But, α is surjective, so there exists an $r \in R$ such that $\alpha(r) = s$. Thus,

$$(\beta\alpha)r = \beta(\alpha(r)) = \beta(s) = t.$$

Since we showed that $\beta\alpha$ is both injective and surjective, it follows that $\beta\alpha$ is bijective. ■

- **Theorem 1.3: Existence of the inverse:** Let $\alpha : S \rightarrow T$ be a bijection. Then, there exists a bijection $\beta : T \rightarrow S$ such that $(\beta\alpha)(s) = s$ for all $s \in S$ and $(\alpha\beta)(t) = t$ for all $t \in T$.

Proof. Since α is bijective, for all $t \in T$, there exist a unique $s \in S$ such that $\alpha(s) = t$. Now, define $\beta : T \rightarrow S$ such that for that unique s that maps to t ,

$$\beta(t) = s.$$

Then,

$$(\beta\alpha)(s) = \beta(\alpha(s)) = \beta(t) = s.$$

Now, for any $t \in T$, choose $s \in S$ such that $\alpha(s) = t$. By that same property of β ,

$$\beta(t) = s,$$

so

$$(\alpha\beta)(t) = \alpha(\beta(t)) = \alpha(s) = t.$$

Now, we just need to show that β is a bijection. First, assume that $\beta(t_1) = \beta(t_2)$. Then,

$$\begin{aligned}\alpha(\beta(t_1)) &= t_1, \\ \alpha(\beta(t_2)) &= t_2.\end{aligned}$$

But, $\beta(t_1) = \beta(t_2)$, so

$$\alpha(\beta(t_2)) = t_2 = \alpha(\beta(t_1)) = t_1.$$

Thus, β is injective. Next, $s \in S$, we have that

$$\beta(\alpha(s)) = s.$$

Thus, there exists a $t \in T$ such that $\beta(t) = s$ ($t = \alpha(s)$), so β is surjective. ■

- **Permutation:** A permutation of a set S is a bijection from S to S .
- **Binary operation:** Let S be a set. Then, a **binary operation** on S is a function from $S \times S$ to S .

3.2 The Integers and Modular Arithmetic

3.2.1 Induction

- **Property 2.1: Well ordering axiom:** If S is a nonempty set of positive integers, then S has a smallest element.
- **Theorem 2.1: Mathematical induction:** Suppose that for each positive integer n , we have a proposition $P(n)$. Further suppose that
 1. $P(1)$ is true, and
 2. for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$

Then, $P(n)$ is true for every positive integer n .

Proof. We proceed by contradiction. Recall that to show a statement $P \implies Q$ is true, it is merely sufficient to show that $P \wedge \neg Q$ is impossible. Thus, if we assume $\neg Q$, and yield a contradiction, $P \wedge \neg Q$ cannot occur and thus the statement is true.

So, we assume that $P(n)$ is not true for every positive integer n . Thus, there is a nonempty set $S \subseteq \mathbb{N}$ such that for every element $n \in S$, $P(n)$ is false. By the well-ordering axiom, there exists a smallest element in S , call this element k .

Observe that the statement assumes that $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$. So, $k > 1$. Further,

$$k-1 \notin S,$$

since if $k-1 \in S$, $k-1+1 = k \notin S$, but k is indeed a member of S . However, since $k-1 \notin S$, $P(k-1)$ is true, and thus $P(k-1+1) = P(k)$ is true, a contradiction.

Therefore, if $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, $P(n+1)$ is also true, then $P(n)$ is true for every positive integer n . ■

Let's consider an example. Suppose we claim that for every positive integer n ,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

First, we observe that

$$1^2 = \frac{1(1+1)(2(1)+1)}{6} = 1.$$

So, $P(1)$ is true. Next, assume that $P(n)$ is true, where

$$P(n) = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Now, observe that

$$\begin{aligned} P(n+1) &= 1^2 + 2^2 + \cdots + n^2 + (n+1)^2 = P(n) + (n+1)^2 \\ &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{n(2n+1)}{6} + (n+1) \\ &= \frac{n(2n+1) + 6(n+1)}{6} = \frac{2n^2 + n + 6n + 6}{6} = \frac{2n^2 + 7n + 6}{6}. \end{aligned}$$

From here, we can get that

$$\frac{2n^2 + 7n + 6}{6} = \frac{(n+2)(2n+3)}{6}.$$

So,

$$\frac{P(n)}{n+1} + (n+1) = \frac{(n+2)(2n+3)}{6}.$$

Thus,

$$P(n) + (n+1)^2 = P(n+1) = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6}$$

Thus, $P(n)$ implies $P(n+1)$, and P therefore holds for every positive integer n .

- **Binomial theorem:** Let a and b be real numbers, and n a positive integer. Then,

$$\begin{aligned} (a+b)^n &= a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}ab^{n-1} + b^n \\ &= \sum_{k=0}^{n-1} \binom{n}{k}a^{n-k}b^k. \end{aligned}$$

Proof. prove by induction on n .

- **Strong induction:** Suppose that, for each positive integer n , we have a proposition $P(n)$. Further suppose that
 1. $P(1)$ is true, and
 2. for each integer $n > 1$, if $P(k)$ is true for every $k < n$, then $P(n)$ is true.

Then, $P(n)$ is true for every positive integer n .

Proof. Assume that the two conditions hold, but $P(n)$ is not true for every positive integer n . Then, let S be the set of integers in which $P(s)$ is false. Since $S \subseteq \mathbb{N}$ and is nonempty, here is a smallest element j by the well-ordering axiom. Call this smallest element ℓ .

Note that since $P(1)$ is true, $\ell > 1$. Since ℓ is the smallest element in S , and therefore the first element that makes the proposition false, it must be that $P(k)$ is true for all $k < \ell$. But, that would imply that $P(\ell)$ is in fact true, a contradiction.

Thus, there is no such set of elements S that make P false, so $P(n)$ true for all $n \in \mathbb{N}$.

3.2.2 Some Number Theory

- **Theorem 2.4 (Division algorithm):** Let $a, b \in \mathbb{Z}$ with $b > 0$. Then, there exists unique integers q and r such that

$$a = bq + r, \quad 0 \leq r < b.$$

We call q and r the **quotient** and **remainder** respectively.

Proof. First, let

$$S = \{a - bt : t \in \mathbb{Z}, a - bt \geq 0\}.$$

First, if $0 \in S$, then $a - bq = 0$ for some $q \in \mathbb{Z}$, and so

$$a = bq + 0$$

and the theorem holds. So, $S \subseteq \mathbb{N}$. We first claim that S is nonempty. Let

$$t = -|a|.$$

Then,

$$a - bt = a + |a|b.$$

If $a \geq 0$, then $a + |a|b \geq 0$ since $b > 0$. If $a \leq 0$ then

$$a + |a|b = a - ab = a(1 - b).$$

Notice that since $b > 0, b \in \mathbb{Z}$, it must be that $b \geq 1$. Thus,

$$1 - b \leq 0.$$

So, $a(1 - b) \geq 0$. Thus, we have shown the existence of at least one t such that $a - bt \in S$, and so S is nonempty.

Since S is a nonempty subset of the natural numbers, we have a least element by the well-ordering axiom, say

$$r = a - bq.$$

Recall that $r \geq 0$ by definition. Suppose that $r \geq b$, then $0 \leq r - b < r$. But,

$$r - b = a - bq + b = a - b(q + 1) < r,$$

a contradiction, since r is the smallest element. Note that

$$r - b = a - b(q + 1)$$

is indeed a member of S , since $r - b \geq 0$. Thus,

$$a = bq + r, \quad 0 \leq r < b.$$

For uniqueness, suppose that

$$a = bq_1 + r_1 = bq_2 + r_2$$

with $q_i, r_i \in \mathbb{Z}$ and $0 \leq r_i < b$. Then,

$$b(q_1 + q_2) = r_1 - r_2.$$

If we take the absolute value,

$$|b(q_1 + q_2)| = |r_1 - r_2|.$$

But, $b > 0$, so

$$b|q_1 - q_2| = |r_2 - r_1|.$$

Notice that since $q_1, q_2 \in \mathbb{Z}$, $q_1 - q_2 \in \mathbb{Z}$, so if $q_1 \neq q_2$,

$$b|q_1 - q_2| \geq b.$$

But, $0 \leq r_1, r_2 < b$, so

$$|r_2 - r_1| < b,$$

and $|r_2 - r_1| = b|q_1 - q_2|$, which we just said was $\geq b$, a contradiction. Hence, $q_1 = q_2$. In that case,

$$b(q_1 - q_2) = b(0) = r_2 - r_1.$$

Thus, $r_1 = r_2$. ■.

- **Divides:** Let a and b be integers, we say that a divides b if there exists an integer c such that $b = ac$, in which case we write

$$a \mid b.$$

If no such c exists, we write $a \nmid b$

- **Lemma 2.1:** Let $a, b, c \in \mathbb{Z}$. Then,
 1. if $a \mid b$ and $b \mid c$, then $a \mid c$
 2. if $a \mid b$ and $b \neq 0$, then $a \leq |b|$
 3. if $a \mid b$ and $a \mid c$, then $a \mid (bu + cv)$ for any $u, v \in \mathbb{Z}$

Proof. First, assume that $a, b, c \in \mathbb{Z}$. (1). Assume that $a \mid b$, and $b \mid c$. Then,

$$b = a\ell, \quad c = bk.$$

If $k = 0$, then $c = 0$, and $a \mid c$. So, assume that $k \neq 0$. Then,

$$b = \frac{c}{k},$$

and so

$$\frac{c}{k} = a\ell.$$

Hence, $c = a(\ell k)$, and $a \mid c$.

(2) Assume that $a \mid b$, $b \neq 0$. So,

$$b = ar, \quad r \in \mathbb{Z}.$$

Consider the case that $a \leq |b|$ is false, so $|b| < a$, or

$$-a < b < a.$$

Notice that $a = 0$ would imply that $b = 0$, but $b \neq 0$. So, $a \neq 0$. Then,

$$-1 < \frac{b}{a} = r < 1.$$

But, this would contradict the fact that $r \in \mathbb{Z}$. So, $a \leq |b|$.

(3). Suppose that $a \mid b$ and $a \mid c$, so

$$b = ar, \quad c = ak, \quad r, k \in \mathbb{Z}.$$

Let $u, v \in \mathbb{Z}$,

$$bu = aru, \quad cv = akv.$$

Then,

$$bu + cv = aru + akv = a(ru + kv).$$

Since $ru + kv \in \mathbb{Z}$, $a \mid bu + cv$ for all $u, v \in \mathbb{Z}$. ■

- **Greatest common divisor:** Let a and b be integers, not both zero. Then, the **greatest common divisor (gcd)** of a and b , written (a, b) , is the largest positive integer g such that $g \mid a$ and $g \mid b$.

Notice that (a, b) is always at least 1, and $(a, b) \leq |a|$. So, only numbers from 1 to $|a|$ are considered.

- **Lemma 2.2:** Take any integers a, b with $a \neq 0$. Then,

1. $(a, b) = (-a, b)$
2. $(a, 0) = |a|$

Proof. Assume a, b are integers with $a \neq 0$.

(1). Assume that $(a, b) = g$, so

$$a = gk, \quad k \in \mathbb{Z}.$$

In that case,

$$-a = g(-k).$$

Since $-k \in \mathbb{Z}$, $g \mid -a$. Trivially, g is still maximal for $(-a, b)$.

(2). Since all numbers divide 0, and $d \leq |a|$ for any number d that divides a , the GCD is therefore $|a|$. ■

- **Relatively prime:** Let $a, b \in \mathbb{Z}$, not both zero. Then, we say that a, b are **relatively prime** if $(a, b) = 1$.
- **Theorem 2.5: Bézout's identity:** Let a and b be integers, not both 0. Then, there exists integers u, v such that

$$(a, b) = au + bv.$$

Furthermore, (a, b) is the smallest positive integer that can be written in this way.

Proof. Let

$$S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\} \subseteq \mathbb{N}.$$

First, we show that S has at least one element. If we assume $a \neq 0$, and let $x = a$, $b = 0$, then

$$a(a) + b(0) = a^2 > 0.$$

So, $S \neq \emptyset$. By the well-ordering axiom, S has a least element, say

$$g = au + bv.$$

If we show that $g = (a, b)$, the proof is complete.

Suppose that a and b has some only common divisor c , then

$$c \mid a \wedge c \mid b.$$

So,

$$c \mid au + bv$$

for all $u, v \in \mathbb{Z}$. Hence, $c \mid g$, and $c \leq g$. That is, g would indeed be the greatest common divisor provided we can show that $g \mid a$ and $g \mid b$.

Using the division algorithm,

$$a = gq + r, \quad 0 \leq r < g.$$

So,

$$r = a - gq = a - (au + bv)q = a - auq - bvq = a(1 - uq) + b(-vq).$$

Hence, $r \in S$ if $r > 0$. But, $r < g$, contradicting the minimality of g . So, $r = 0$, and $g \mid a$. Using the same argument, we can also get that $g \mid b$. ■

- **Corollary 2.1:** Let $a, b \in \mathbb{Z}$, $\{a, b\} \neq \{0\}$. Then, a and b are relatively prime if and only if there exists integers u, v such that

$$au + bv = 1.$$

- **Corollary 2.2:** Let $a, b, c \in \mathbb{Z}$ with $\{a, b\} \neq \{0\}$. If $(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Proof. We have

$$au + bv = 1, \quad u, v \in \mathbb{Z}.$$

So,

$$auc + bvc = c.$$

Since $a \mid a$ and $a \mid bc$,

$$a \mid a(uc) + bc(v),$$

since $uc, v \in \mathbb{Z}$. Thus, $a \mid c$. ■

- **Corollary 2.3:** Let $a, b \in \mathbb{Z}$, $\{a, b\} \neq \{0\}$. If $(a, b) = 1$, and for same integer n we have $a \mid n$ and $b \mid n$, then $ab \mid n$.
- **Theorem 2.6: Euclidean algorithm:** Let a, b be integers, with b positive. If $b \mid a$, then $(a, b) = b$. Otherwise, apply the division algorithm repeatedly. Let

$$\begin{aligned} a &= bq_1 + r \\ b &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ &\vdots \\ r_{k-2} &= r_{k-1}q_k + r_k \\ r_{k-1} &= r_kq_{k+1} + 0, \end{aligned}$$

where $q_i, r_i \in \mathbb{Z}$ for all i , and $0 < r_k < r_{k-1} < \dots < r_1 < b$. Then, $(a, b) = r_k$.

Note: You may have seen this algorithm defined recursively as

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Proof. If $b \mid a$, then $b \leq |a|$. Note that $b \mid b$, so b is trivially the greatest divisor of b . Thus, since $b \leq a$ and b is the greatest divisor of b , it should be clear that $(a, b) = b$. Thus, we may assume that $b \nmid a$.

By the division algorithm,

$$a = bq_1 + r_1, \quad q_1, r_1 \in \mathbb{Z}.$$

Notice that $r_1 = a - bq_1$. Assume $c \mid a$ and $c \mid b$. Then,

$$c \mid a(1) + b(-q_1) = a - bq_1 = r_1.$$

Thus, any common divisor of a and b is also a common divisor of b and r_1 , with $r_1 = a \bmod b$. Conversely, if $d \mid b$ and $d \mid r_1$, then

$$d \mid b(q_1) + r_1(1) = a.$$

So, any common divisor of b and r_1 is also a common divisor of a and b . Hence, $b, a \bmod b$ have precisely the same common divisors as a and b .

Also, observe that

$$0 < r_1 < b.$$

If $r_1 = 0$, then $b \mid a$, so $r_1 > 0$. Now, we can repeat this argument to say that the common divisors of b, r_1 are the same as the common divisors of r_1, r_2 , where $r_2 = b \bmod r_1 = b \bmod a \bmod b$, with

$$0 < r_2 < r_1.$$

Thus,

$$(a, b) = (b, r_1) = (r_1, r_2) = \cdots = (r_{k-1}, r_k) = (r_k, 0) = r_k.$$

Since $0 < r_k < r_{k-1} < \cdots < r_2 < r_1 < b$, each iteration produces a smaller r_i than the one prior, and since all are greater than zero, the sequence must terminate in finite steps.

Note: We note that if $b < 0$, we cannot use the division algorithm, as that requires $b > 0$. This is why we assume b is positive for this theorem. However, recall that

$$(a, b) = (-a, b).$$

This fact, along with $(a, b) = (b, a)$ lets us use the algorithm with

$$(a, -b)$$

to find the gcd of (a, b) .

- **Using the Euclidean algorithm:** Note that since

$$r_{k-2} = r_{k-1}q_k + r_k \implies (a, b) = r_k = r_{k-2} - r_{k-1}q_k = r_{k-2}(1) + r_{k-1}(-q_k),$$

with

$$\begin{aligned}
a &= bq_1 + r_1, \\
b &= r_1q_2 + r_2, \\
r_1 &= r_2q_3 + r_3, \\
&\vdots \\
r_{k-4} &= r_{k-3}q_{k-2} + r_{k-2}, \\
r_{k-3} &= r_{k-2}q_{k-1} + r_{k-1},
\end{aligned}$$

we can work our way backwards to get integers u, v such that

$$(a, b) = au + bv.$$

For example, consider $a = 45$, $b = 33$. Applying the Euclidean algorithm yields

$$\begin{aligned}
45 &= 33(1) + 12, \\
33 &= 12(2) + 9, \\
12 &= 9(1) + 3, \\
9 &= 3(3) + 0.
\end{aligned}$$

Hence, $(a, b) = 3$. Now, we can find u, v such that $au + bv = 3$. We have

$$\begin{aligned}
3 &= r_1(1) + r_2(-q_3) = 12(1) + 9(-1), \\
&= r_1(1) + (b - r_1q_2)(-q_3) = 12(1) + (33 - 12(-2))(-1) \\
&= (a - bq_1) + b(-q_3) + r_1q_2q_3 \\
&= (a - bq_1) + b(-q_3) + (a - bq_1)q_2q_3 \\
&= a + b(-q_1) + b(-q_3) + aq_2q_3 + b(-q_1)q_2q_3 \\
&= a(1 + q_2q_3) + b(-q_1 - q_3 - q_1q_2q_3) \\
&= 45(1 + 2(1)) + 33(-1 - 1 - 1(2)(1)) = 45(3) + 33(-1 - 1 - 1(2)(1)) = 45(3) + 33(-4).
\end{aligned}$$

Hence, $u = 3$, and $v = -4$.

- **Prime numbers:** A natural number $p > 1$ is said to be prime if its only positive divisors are 1 and p . Otherwise, it is composite. Note that 1 is neither prime nor composite.
- **Euclid's Lemma:** Let $p > 1$ be a positive integer. Then, the following are equivalent.
 1. p is prime; and
 2. if a and b are integers such that $p \mid ab$, then $p \mid a$ or $p \mid b$

Proof. First, suppose that p is prime and $p \mid ab$. Either $(p, a) = 1$ or $(p, a) = p$. If $p \leq a$, we have $(p, a) = p$, and so p is a common divisor of both p and a , hence $p \mid a$.

If $p > a$, then $(p, a) = 1$. In that case,

$$pu + av = 1,$$

so

$$pub + avb = b.$$

Now, with $p \mid p$ and $p \mid ab$,

$$p \mid pk + abc, \quad k, c \in \mathbb{Z}.$$

If we let $k = ub$, $c = v$, then

$$p \mid p(ub) + abv = b.$$

So, $p \mid b$. Now, consider the case in which p is composite. In that case, let $p = cd$. Since $p > 1$,

$$1 < c, d < p.$$

So, $p \nmid c$ and $p \nmid d$. ■

- **Corollary 2.4:** Let p be a prime number and $a_1, \dots, a_n \in \mathbb{Z}$. If

$$p \mid a_1 a_2 \cdots a_n,$$

then $p \mid a_i$ for some i . In fact, every positive integer larger than 1 can be written as a product of primes, called its **prime factorization**.

Proof. Assume $p \in \mathbb{P}$, and $a_1, a_2, \dots, a_n \in \mathbb{Z}$. Further assume that $p \mid a_1 a_2 \cdots a_n$.

So, $p \mid a_1$ or $p \mid a_2 \cdots a_n$. Further, $p \mid a_2$ or $p \mid a_3 \cdots a_n$. We can extend this idea up to $p \mid a_{n-1}$ or $p \mid a_n$. Thus, $p \mid a_i$ for some i . ■

- **Theorem 2.8: Fundamental Theorem of Arithmetic:** If $a \in \mathbb{N}$ and $a > 1$, then there exists primes $p_1, \dots, p_n$ not necessarily distinct such that $a = p_1 p_2 \cdots p_n$. Further, this product is unique up to order. That is, if $a = q_1 q_2 \cdots q_m$ for some primes q_i , then $m = n$, and after rearranging the primes, $p_i = q_i$ for all i .

Proof. Notice that we start with $a = 2$, which is prime. Thus, let $a > 2$ and assume that the theorem is true for smaller numbers. If a is prime, nothing to do. If a is composite, then we can write $a = bc$ for $1 < b, c < a$. Since $b, c < a$, b and c can be written as a product of primes. Thus, a is a product of primes.

Now we prove uniqueness. Suppose that

$$a = p_1 \cdots p_n = q_1 \cdots q_m$$

for some primes p_i and q_i . Without loss of generality, say $n \leq m$. Since

$$p_1 \mid a,$$

$p_1 \mid q_i$ for some i . We can rearrange the order to say that $p_1 \mid q_1$. But, q_1 is prime, so $p_1 = 1$ or $p_1 = q_1$. Since 1 is not prime, $p_1 = q_1$. So, we can cancel p_1 from both sides to get

$$p_2 \cdots p_n = q_2 \cdots q_m.$$

We can continue this process, finding that $p_i = q_i$ for $1 \leq i \leq n$. If $m = n$, we are done. Otherwise, we are left with $1 = q_{n+1} \cdots q_m$. But then $q_m \mid 1$, which is impossible since $q_m > 1$. ■

- **Corollary 2.5:** Let a, b , and n be integers, with $n \neq 0$. If $(a, n) = (b, n) = 1$, then $(ab, n) = 1$.

Proof. Assume that $a, b, n \in \mathbb{Z}$, $n \neq 0$, $(a, n) = (b, n) = 1$.

Assume that $g = (ab, n) > 1$. By the Fundamental Theorem of Arithmetic, g can be factorized as a product of primes. Hence, there exists a prime p that divides g . Since g divides ab and p divides g , $p \mid ab$. In that case, $p \mid a$ or $p \mid b$. But, $p \mid n$ by similar logic. So,

$$(a, n) \geq p \quad \text{or} \quad (b, n) \geq p.$$

Either way, we have a contradiction, since $(a, n) = (b, n) = 1$. ■.

3.2.3 Properties of the integers

- **Intro:** We note that our discussion will take place in $\mathbb{Z}$, but it could just as easily use $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$.

- **Closure:** $\mathbb{Z}$ is closed under addition and multiplication

$$a + b, ab \in \mathbb{Z}$$

for all $a, b \in \mathbb{Z}$.

- **Associativity:** Addition and multiplication on $\mathbb{Z}$ are both associative.

$$\begin{aligned}(a + b) + c &= a + (b + c), \\ (ab)c &= a(bc)\end{aligned}$$

for all $a, b, c \in \mathbb{Z}$.

- **Commutative:** Addition and multiplication are both commutative on $\mathbb{Z}$.

$$\begin{aligned}a + b &= b + a, \\ ab &= ba\end{aligned}$$

for all $a, b \in \mathbb{Z}$.

- **Identity:** We call 0 the additive identity for $\mathbb{Z}$, and 1 the multiplicative identity for $\mathbb{Z}$

$$a + 0 = a, \quad 1a = a$$

for all $a \in \mathbb{Z}$.

- **Inverse:** If $a \in \mathbb{Z}$, then $-a$ is its additive inverse.

$$a + (-a) = 0.$$

We note that there is no multiplicative inverse in $\mathbb{Z}$, unlike $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$.

3.2.4 Modular arithmetic

- **Intro:** Consider $a, b, n \in \mathbb{Z}$. By the division algorithm,

$$\begin{aligned} a &= nc + r_1, \\ b &= nk + r_2 \end{aligned}$$

for $c, k, r_1, r_2 \in \mathbb{Z}$, $0 \leq r_1, r_2 < n$. Now, let's consider $a - b$,

$$a - b = nc + r_1 - nk - r_2 = n(c - k) + (r_1 - r_2).$$

Notice that if $n \mid a - b$, then $r_1 - r_2 = 0$, so $r_1 = r_2$. Thus, if $n \mid a - b$, a and b have the same remainder when divided by n .

- **Congruence:** Let $n \geq 2$ be an integer. If $a, b \in \mathbb{Z}$, then we say that a is congruent to b modulo n , and write $a \equiv b \pmod{n}$ provided

$$n \mid a - b.$$

That is, a and b have the same remainder when divided by n .

- **Reduction to the remainder:** Suppose that $a, b, n \in \mathbb{Z}$, $n \geq 2$, and $n \mid a - b$. So,

$$a \equiv b \pmod{n}.$$

In that case, a and b have the same remainder when divided by n , so

$$\begin{aligned} a &= nc + r, \\ b &= nk + r \end{aligned}$$

for $c, k \in \mathbb{Z}$. Notice

$$\begin{aligned} a - r &= nc, \\ b - r &= nk. \end{aligned}$$

So, $n \mid a - r$, and $n \mid b - r$. Thus, we can reduce to

$$\begin{aligned} a &\equiv r \pmod{n}, \\ b &\equiv r \pmod{n}. \end{aligned}$$

- **Theorem 2.9:** Let $n \geq 2$ be an integer. Then, $a \equiv b \pmod{n}$ is an equivalence relation on $\mathbb{Z}$. The equivalence class of a consists of all integers having the same remainder as a when divided by n .

Proof. Assume $n \geq 2$ is an integer, let $a \equiv b \pmod{n}$ be a relation on $\mathbb{Z}$. First, observe that

$$n \mid 0 = a - a.$$

Thus, $a \equiv a \pmod{n}$, and the relation is reflexive. Next, Assume that

$$a \equiv b \pmod{n}.$$

Thus, $n \mid a - b$, or

$$a - b = nc$$

for some $c \in \mathbb{Z}$. Then,

$$-(b - a) = nc \implies b - a = n(-c).$$

Thus, $n \mid b - a$, so $b \equiv a \pmod{n}$ and the relation is symmetric. Last, assume that

$$a \equiv b \pmod{n}, \quad b \equiv c \pmod{n}.$$

Then,

$$a - b = nr, \quad b - c = nk$$

for some $r, k \in \mathbb{Z}$. Thus,

$$a - b + b - c = a - c = nr + nk = n(r + k).$$

Thus, $n \mid a - c$, so $a \equiv c \pmod{n}$ and the relation is transitive. Hence, we have an equivalence relation on $\mathbb{Z}$.

And, as we know for an equivalence relation $\sim \subseteq S \times S$,

$$[a] = \{b \in S : a \sim b\},$$

and if $b \in [a]$, $[a] = [b]$. Thus, for the equivalence relation $a \equiv b \pmod{n}$, let $a \in \mathbb{Z}$, then the equivalence class $[a]$ is

$$[a] = \{b \in \mathbb{Z} : a \equiv b \pmod{n}\} = \{a + nk : k \in \mathbb{Z}\}.$$

So, the integers $b \in \mathbb{Z}$ with the same remainder when divided by n makes up the equivalence class for a . ■

- **Definition 2.6:** Let $n \geq 2$ be an integer. The set of integers modulo n , denoted $\mathbb{Z}_n$ is the set of equivalence classes of $\mathbb{Z}$ with respect to the equivalence relation $a \equiv b \pmod{n}$. We call these the **congruence classes** modulo n . Specifically,

$$\mathbb{Z}_n = \{[0], [1], [2], \dots, [n-1]\}.$$

Recall that the representative of a class is not unique. However, it is customary to reduce final answers in $\mathbb{Z}_n$ to the form $[a]$, where $0 \leq a < n$.

- **Residue class:** The term **residue class** is another name for **congruence class**. A **residue** is usually a representative of that class.
- **Sumsets:** Consider $S, T \subseteq \mathbb{Z}$. Then,

$$S + T = \{s + t : s \in S, t \in T\}$$

is called the **sumset**. Similarly, we can define the **difference set**, **product set**, or **quotient set** in a similar way, just changing the operation.

- **Addition and multiplication on $\mathbb{Z}_n$:** We have

$$\begin{aligned} [a] + [b] &= [a + b], \\ [a][b] &= [ab]. \end{aligned}$$

Consider the sumset

$$[a] + [b] = \{x + y : x \in [a], y \in [b]\}.$$

Note that

$$[a + b] = \{s \in \mathbb{Z} : a + b \equiv s \pmod{n}\}.$$

So, we are asserting that

$$\{x + y : x \in [a], y \in [b]\} = \{s \in \mathbb{Z} : a + b \equiv s \pmod{n}\}.$$

Proof. Let $x + y = [a] + [b]$, so $x \in [a]$, and $y \in [b]$. Hence,

$$\begin{aligned} a &\equiv x \pmod{n}, \\ b &\equiv y \pmod{n}. \end{aligned}$$

Thus,

$$a - x = nc, \quad b - y = nk$$

for some $c, k \in \mathbb{Z}$. Thus,

$$a - x + b - y = a + b - (x + y) = n(c + k).$$

Hence,

$$a + b \equiv x + y \pmod{n}.$$

So, $x + y \in [a + b]$, and $[a] + [b] \subseteq [a + b]$. Now, let $z \in [a + b]$, so

$$z \equiv a + b \pmod{n}.$$

Let $x = a$, and $y = z - a$. Then, $x + y = z$,

$$x \equiv a \pmod{n},$$

and

$$z - a = y \equiv (a + b) - a \pmod{n}.$$

So, $y \equiv b \pmod{n}$, hence $x \in [a]$ and $y \in [b]$. Therefore $[a + b] \subseteq [a] + [b]$, and we can conclude that

$$[a] + [b] = [a + b].$$

■

Now, consider

$$\begin{aligned} [a][b] &= \{xy : x \in [a], y \in [b]\}, \\ [a \cdot b] &= \{z \in \mathbb{Z} : ab \equiv z \pmod{n}\}. \end{aligned}$$

We assert that

$$\{xy : x \in [a], y \in [b]\} = \{z \in \mathbb{Z} : ab \equiv z \pmod{n}\}.$$

The proof for this is non-trivial. Instead of proving these two operations, we can instead define them as such, and show that they are well-defined.

- **A well-defined operation on a set:** An operation is well-defined on a set if it gives a single, unambiguous output in that set for every allowed input from that set. For a binary operation $*$ on a set S ,

$$* : S \times S \rightarrow S.$$

$*$ is well-defined on S if for every $a, b \in S$, $a * b$ exists, is unique, and is also an element of S .

- **Checking well-definedness:** You must show well-definedness whenever your definition uses representatives of equivalence classes. As well as for defining operations on a set.

In general, whenever you define an operation on a set, you should at least know that it is well-defined. You should check well-definedness especially when the rule involves choices, representatives, cases, or possible ambiguity.

- **Theorem 2.10:** For any integer $n \geq 2$, addition and multiplication on $\mathbb{Z}_n$ are well-defined.

Proof. Suppose that $a_1 \equiv a_2 \pmod{n}$ and $b_1 \equiv b_2 \pmod{n}$. Then,

$$\begin{aligned} a_1 - a_2 &= nc, \\ b_1 - b_2 &= nk. \end{aligned}$$

So,

$$a_1 + b_1 - (a_2 + b_2) = n(c + k).$$

Thus,

$$a_1 + b_1 \equiv a_2 + b_2 \pmod{n}.$$

Hence, $[a_1 + b_1] = [a_2 + b_2]$, and we can say that addition is well-defined.

Now,

$$\begin{aligned} a_1 &= nc + a_2, \\ b_1 &= nk + b_2. \end{aligned}$$

So,

$$a_1 b_1 = (nc + a_2)(nk + b_2) = n^2 ck + ncb_2 + nka_2 + a_2 b_2.$$

Hence,

$$a_1 b_1 - a_2 b_2 = n(nck + cb_2 + ka_2),$$

and we can say that $a_1 b_1 \equiv a_2 b_2$, so

$$[a_1 b_1] = [a_2 b_2].$$

Therefore, multiplication on residue classes is also well-defined.

- **Theorem 2.11:** Let $n \geq 2$ be an integer, and take any $[a], [b], [c] \in \mathbb{Z}_n$. Then,

1. $[a] + [b] \in \mathbb{Z}_n$
2. $[a] + ([b] + [c]) = ([a] + [b]) + [c]$
3. $[a] + [b] = [b] + [a]$
4. $[a] + [0] = [a]$
5. $[a] + [-a] = [0]$

- **Theorem 2.12:** Let $n \geq 2$ be an integer, and $[a], [b], [c] \in \mathbb{Z}_n$, then

1. $[a][b] \in \mathbb{Z}_n$
2. $[a]([b][c]) = ([a][b])[c]$
3. $[a][b] = [b][a]$
4. $[a]([b] + [c]) = [a][b] + [a][c]$
5. $[1][a] = [a]$

- **Notational switch:** When working in $\mathbb{Z}_n$, we will normally simply write a or $a + b$ instead of $[a]$ or $[a] + [b]$, as long as the context is clear.

3.3 Groups

3.3.1 Introduction to groups

- **Permutations:** Let A be the set $\{1, 2, 3\}$. We would like to consider all the permutations of A . The notation we will use is

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ a & b & c \end{pmatrix},$$

which means $\sigma(1) = a, \sigma(2) = b$, and $\sigma(3) = c$.

- **Composition of permutations:** Consider

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Then,

$$\begin{aligned} (\sigma \circ \tau)(1) &= \sigma(\tau(1)) = \sigma(3) = 3, \\ (\sigma \circ \tau)(2) &= \sigma(\tau(2)) = \sigma(1) = 2, \\ (\sigma \circ \tau)(3) &= \sigma(\tau(3)) = \sigma(2) = 1. \end{aligned}$$

Thus,

$$\sigma \circ \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

- **Properties of permutations:**

1. **Closure:** Permutations are closed under composition
2. **Associativity:** For permutations σ, τ, ρ , we have $\rho \circ (\sigma \circ \tau) = (\rho \circ \sigma) \circ \tau$
3. **Identity:** If σ is any permutation of A , then

$$\sigma \circ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \circ \sigma = \sigma.$$

4. **Inverse:** For each permutation σ , there is another permutation τ such that

$$\sigma \circ \tau = \tau \circ \sigma = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Note that we do not have commutativity, $\sigma \circ \tau \neq \tau \circ \sigma$ in general.

- **Symmetric group on n letters, S_n :** Permutations under composition give us an example of a group. If we let $A = \{1, 2, 3, \dots, n\}$, for any positive integer n , then we call the set of all permutations of this set, under the composition operation the **symmetric group on n letters**, denoted S_n .
- **Groups:** A **group** is a set G , together with a binary operation $*$, satisfying the following conditions
 1. $a * b \in G$ for all $a, b \in G$ (closure);
 2. $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$ (associativity);
 3. there exists an $e \in G$ such that $a * e = e * a = a$ for all $a \in G$ (existence of identity); and
 4. for each $a \in G$, there exists a $b \in G$ such that $a * b = b * a = e$ (existence of inverses).

We refer to G as a group **under** $*$. The element e is called the **identity** of the group. Also, if b is the inverse of a , then we write $b = a^{-1}$.

- **Abelian group:** A group G is said to be **abelian** if $a * b = b * a$ for all $a, b \in G$. So, $*$ is commutative.

Note that $\mathbb{Z}$ and $\mathbb{Z}_n$ for $n \geq 2$ are abelian groups under addition. In fact, 0 is the identity, and the inverse of a is $-a$. The same can be said for $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ under addition.

- **Group tables:** When a group G has only finitely many elements, we can represent it with a **group table**. We write the elements of G down the first column and along the first row of the table. Then, the entry in the row headed by a and the column headed by b is $a * b$. Observe the group table for the additive group $\mathbb{Z}_5$

	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

- **Example:** Let G be the set of nonzero rational numbers. Then, G is an abelian group under multiplication.

Proof. Let

$$G = \{q \in \mathbb{Q} : q \neq 0\} \subseteq \mathbb{Q}.$$

Consider $q_1, q_2, q_3 \in G$, with

$$q_1 = \frac{a}{b}, q_2 = \frac{c}{d}, q_3 = \frac{e}{f}.$$

Then,

$$q_1 q_2 = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

It should be clear that $a, b, c, d \neq 0$, so $q_1 q_2 \in G$. Thus, we have closure. Next,

$$q_1 \cdot (q_2 \cdot q_3) = (q_1 \cdot q_2) \cdot q_3$$

since multiplication is associative. Also,

$$q_1 q_2 = q_2 q_1$$

since multiplication is commutative. Clearly $e = 1$ is the identity, since $1 \in G$ and $1q = q$ for all $q \in G$. Also, for

$$q = \frac{a}{b} \in G,$$

$$q \cdot \frac{1}{q} = \frac{a}{b} \cdot \frac{b}{a} = 1.$$

Thus, $q^{-1} = \frac{1}{q}$.

- **Example:** Let $n \geq 2$ be positive integer. Let $U(n)$ denote the set of all elements $a \in \mathbb{Z}_n$ such that $(a, n) = 1$.

So,

$$U(n) = \{a \in \mathbb{Z}_n : (a, n) = 1\}.$$

First, let's check that the condition $(a, n) = 1$ is well-defined. Thus, if

$$a \equiv b \pmod{n}$$

with $(a, n) = 1$, we should see that $(b, n) = 1$. Well, we have

$$\begin{aligned} a &= b + nk, \\ au + nv &= 1. \end{aligned}$$

So,

$$(b + nk)u + nv = bu + nku + nv = bu + n(ku + v) = 1.$$

Thus, $(b, n) = 1$. Alternately, since a is a linear combination of b and n , then if we let c be a divisor of b and n , we can invoke lemma 2.1 to get that

$$c \mid b(1) + nk.$$

So, $c \mid a$. Now, c divides both a and n , but the only common divisor for a and n is 1. Thus, $c = 1$. Now, notice that we choose an arbitrary divisor for b and n , and deduced that it must be one. Thus, b and n have no common divisor other than 1, so

$$(b, n) = 1.$$

■.

Now that we have well-definedness, we can claim that $U(n)$ forms an abelian group under multiplication in $\mathbb{Z}_n$.

First, we check for closure. If $a, b \in U(n)$, then we should see that $ab \in U(n)$. So, we require that

$$(ab, n) = 1.$$

In fact, we see this to be true by Corollary 2.5. We also know multiplication in $\mathbb{Z}_n$ to be both associative and commutative, and $1 \in U(n)$ is the identity. But, what about inverses? For a residue class a , we call $[b] = a^{-1}$ if

$$ab \equiv 1 \pmod{n}.$$

In other words,

$$ab = 1 + nk, \quad k \in \mathbb{Z}.$$

Notice that since $(a, n) = 1$,

$$au + nv = 1 \implies au = 1 - nv = 1 + nk.$$

Thus, $u = a^{-1}$.

- **Example:** Let n be a positive integer, and

$$(G, \cdot) = (\{w \in \mathbb{C} : w^n = 1\}, \cdot).$$

If $w, z \in G$, then

$$(wz)^n = w^n z^n = 1(1) = 1.$$

So, we have closure. Next, we know that multiplication is both associative and commutative, and $1 \in G$ is the identity. Now, for $w \in G$, $\frac{1}{w} \in \mathbb{C}$, and

$$\left(\frac{1}{w}\right)^n = \frac{1}{w^n} = \frac{1}{1} = 1.$$

So, $\frac{1}{w} \in G$, and $\frac{1}{w} = w^{-1}$, since

$$w \left(\frac{1}{w}\right) = 1.$$

- **Trivial group:** We call the group that consists of just the identity element the **trivial group**.
- **General linear group:** The set of all non-singular matrices in $M_n(\mathbb{R})$ ($\mathbb{R}^{n \times n}$) is called the **general linear group** over $\mathbb{R}$, and denoted $GL_n(\mathbb{R})$. It is a group under matrix multiplication. The identity matrix I_n is the identity of $GL_n(\mathbb{R})$. Also, if $A, B \in GL_n(\mathbb{R})$, then

$$AB(B^{-1}A^{-1}) = A(BB^{-1})A = I_n,$$

and

$$(B^{-1}A^{-1})AB = I_n.$$

So, we have closure. Also, we know that matrix multiplication is associative, and by definition every element of A has an inverse. Notice that

$$(A^{-1})^{-1} = A,$$

so $A^{-1} \in GL_n(\mathbb{R})$. Thus, this is indeed a group. However, we know that matrix multiplication is not generally commutative, so the group is nonabelian.

- **Definition 3.3 (direct product):** Let G be a group with operation $*$ and H group with operation $\bullet$. On the Cartesian product $G \times H$, define an operation $\diamond$ via

$$(g_1, h_1) \diamond (g_2, h_2) = (g_1 * g_2, h_1 \bullet h_2)$$

for all $g_i \in G$, $h_i \in H$. Under this operation, we call $G \times H$ the **direct product** of G and H .

Note: So, the underlying set is the Cartesian product, with operation $\diamond$. Thus, the group is

$$(G \times H, \diamond).$$

- **Theorem 3.1:** The direct product of two groups is a group.

Proof. Assume that $(G, *)$ and $(H, \bullet)$ are groups. Then, let $(G \times H, \diamond)$ be the direct product.

If $a_1, a_2 \in G \times H$, then $a_1 = (g_1, h_1)$, and $a_2 = (g_2, h_2)$. Now,

$$a_1 \diamond a_2 = (g_1, h_1) \diamond (g_2, h_2) = (g_1 * g_2, h_1 \bullet h_2).$$

Since G, H are groups, $g_1 * g_2 \in G$, and $h_1 \bullet h_2 \in H$, so

$$(g_1 * g_2, h_1 \bullet h_2) \in G \times H.$$

Thus, we have closure. Now, let $a_1, a_2, a_3 \in G \times H$, we have

$$\begin{aligned} ((g_1, h_1) \diamond (g_2, h_2)) \diamond (g_3, h_3) &= (g_1 * g_2, h_1 \bullet h_2) \diamond (g_3, h_3) \\ &= (g_1 * g_2 * g_3, h_1 \bullet h_2 \bullet h_3), \end{aligned}$$

with

$$\begin{aligned} (g_1, h_1) \diamond ((g_2, h_2) \diamond (g_3, h_3)) &= (g_1, h_1) \diamond (g_2 * g_3, h_2 \bullet h_3) \\ &= (g_1 * g_2 * g_3, h_1 \bullet h_2 \bullet h_3). \end{aligned}$$

So, we have associativity. If $e_G \in G$, and $e_H \in H$ are the identities for G and H , then its painfully obvious that

$$(e_G, e_H)$$

is the identity for $G \times H$. Furthermore, (g^{-1}, h^{-1}) is the inverse. ■

3.3.2 A few basic properties

- **Notation change:** From now on, when working inside a group, we will suppress the symbol for the group operation. So, we will write ab instead of $a * b$. Although, if the operation is addition, the **multiplicative notation** would be confusing. In that case, we will use the **additive notation**, and continue with $a + b$

Further, if the operation is addition, we will write 0 instead of e , and $-a$ instead of a^{-1} .

- **Theorem 3.2:** Let G be a group. Then,
 1. the identity of G is unique, and
 2. if $a \in G$, then a^{-1} is unique.

Proof. (1) Suppose that G is a group, and e, f are identities in G . Then,

$$\begin{aligned} ef &= f, \\ ef &= e. \end{aligned}$$

Thus, $e = f$. Now, suppose that for $a \in G$, b, c are both inverses for a . Then,

$$\begin{aligned} bac &= (ba)c = ec = c, \\ bac &= b(ac) = be = b. \end{aligned}$$

Thus, $c = b$.

- **Theorem 3.3:** In any group, $(ab)c = a(bc)$. Thus, we can write abc without any ambiguity. But, we would like to be able to write $abcd$, for instance. This brings us to the following theorem

Let G be any group, and $a_1, a_2, \dots, a_n \in G$. Then, regardless of how the product $a_1 a_2 \cdots a_n$ is bracketed, the result equals

$$(\cdots(((a_1 a_2) a_3) a_4) \cdots a_{n-1}) a_n.$$

Proof. We proceed by strong induction upon n . If n is 1 or 2, no bracketing is needed. If $n = 3$, we have associativity from the definition of a group. So, let $n \geq 4$. Assume that the theorem is true for any product of fewer than n group elements.

Let

$$w = a_1 \cdots a_n$$

be a product with any bracketing configuration. Then, let

$$w = xy$$

for $x = a_1 \cdots a_m$, $y = a_{m+1} \cdots a_n$, each with some bracketing. By the inductive hypothesis,

$$\begin{aligned} x &= (\cdots(((a_1 a_2) a_3) \cdots a_{m-1}) a_m, \\ y &= (\cdots(((a_{m+1} a_{m+2}) a_{m+3}) \cdots a_{n-1}) a_n. \end{aligned}$$

So,

$$w = xy = ((\cdots(((a_1 a_2) a_3) \cdots a_m) (\cdots(((a_{m+1} a_{m+2}) a_{m+3}) \cdots a_{n-1})) a_n.$$

Applying our inductive hypothesis to the product of the first $n - 1$ terms, we obtain the desired bracketing. ■

- **Theorem 3.4:** Let G be a group, with $a, b \in G$. Then,

1. $(a^{-1})^{-1} = a$
2. $(ab)^{-1} = b^{-1}a^{-1}$

Proof. Since

$$aa^{-1} = a^{-1}a = e,$$

$(a^{-1})^{-1} = a$ by definition of the inverse. Next,

$$\begin{aligned} abb^{-1}a^{-1} &= aea^{-1} = aa^{-1} = e, \\ b^{-1}a^{-1}ab &= b^{-1}eb = b^{-1}b = e. \end{aligned}$$

Thus, $(ab)^{-1} = b^{-1}a^{-1}$. ■

- **Theorem 3.5 (cancellation law):** Let G be a group and $a, b, c \in G$. If either $ab = ac$ or $ba = ca$, then $b = c$.

Proof. Let $ab = ac$, then

$$a^{-1}ab = aa^{-1}ac \implies b = c.$$

Similarly, if $ba = ca$, then

$$baa^{-1} = caa^{-1} \implies b = c. \quad \text{■}$$

- **Corollary 3.1:** Let G be a group, and $a, b \in G$. Then, there exists exactly one $c \in G$ such that $ac = b$, and exactly one $d \in G$ such that $da = b$.

Proof. Let $a, b \in G$. Assume that there exists two elements $c, f \in G$ such that

$$ac = af = b.$$

Then, by Theorem 3.5,

$$ac = af \implies c = f.$$

Now, assume that $d, f \in G$ such that

$$da = fa = b.$$

Then,

$$da = fa \implies d = f. \quad \text{■}$$

3.3.3 Powers and orders

- **Definition 3.4:** If G is a group, then its **order**, $|G|$, is the number of elements in the set G . We say that G is a **finite group** if its order is finite; otherwise it is an **infinite group**.
- **Powers of group elements:** Let G be any group, and $a \in G$. Then, for any positive integer n , we let

$$a^n = \underbrace{aa \cdots a}_{n \text{ times}}.$$